Sources share one 10-minute grid, are routed down the river, then decoded

Station levels
S1–S4 and T1
[B, 48, 5]

Rainfall
R1, R2, mm per slot
[B, 48, 2]

Past releases
actual rate, deviation
[B, 48, 2]

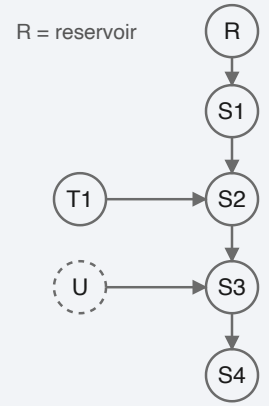
Announced releases
next 6 slots + flag
[B, 6, 2]

1. Align to the 10-min grid: 48 slots ending at τ0
levels: last reading per slot · rain: sum per slot · releases: mean rate
adds a mask m (reading present?) and age a (time since last valid)

Release encoder
known future input
[B, 6, 2] → [B, 6, 32]

2. Project onto the river graph → X [B, 48, 7, 15]
station series → its node; release → node R; rain, calendar, static → all
U (unmonitored tributary) has no level: mask 0, rain features only

River graph



3. Per-node temporal encoder (shared GRU) → H [B, 48, 7, 64]
causal: slot t uses only slots up to t; same weights for every node
a node-type embedding tells the GRU which node it is reading

4. River routing, upstream to downstream → Z [B, 48, 7, 64]
each edge u→v sends u's state from 0–24 slots back (0–4 h)
lag weights depend on flow; keep τ0 at S1–S4 → [B, 4, 64]

5. Decoder per station and horizon → Ŷ [B, 6, 4, 3]
ŷ = last valid level + MLP(station state, release context up to h, horizon embedding)
6 horizons (10–60 min) × stations S1–S4 × quantiles q10, q50, q90